

Integration - Set 2 - Solutions

Specification MM7.1-MM7.6.

Answer key

Question	Answer	Topic
1	D	Integration
2	D	Integration
3	B	Integration
4	F	Integration
5	E	Integration
6	E	Integration
7	A	Integration
8	D	Integration
9	E	Integration
10	D	Integration

Worked solutions

Question 1 INT-18 · Integration · spec MM7.3

Let $I = \int_0^1 (x^2 + 1) dx$ and $J = \int_0^1 (x + 1)^2 dx$. Which of the following is true?

Answer: D

$$I = \frac{1}{3} + 1 = \frac{4}{3}, J = \int_0^1 (x^2 + 2x + 1) dx = \frac{1}{3} + 1 + 1 = \frac{7}{3}, J - I = 1.$$

Full method - how and why

Two similar-looking integrals. Expand the second - $(x + 1)^2 \neq x^2 + 1$ - and compare.

1 Compute I : $\int_0^1 (x^2 + 1) dx = \left[\frac{x^3}{3} + x \right]_0^1 = \frac{1}{3} + 1 = \frac{4}{3}$.

why: Term by term.

2 Expand the second integrand: $(x + 1)^2 = x^2 + 2x + 1$.

why: The cross term $2x$ is the whole difference between the two integrals.

3 Compute J : $\left[\frac{x^3}{3} + x^2 + x \right]_0^1 = \frac{1}{3} + 1 + 1 = \frac{7}{3}$.

why: The extra $\int_0^1 2x dx = 1$.

4 Compare: $J - I = \frac{7}{3} - \frac{4}{3} = 1$, and $J > I$.

why: Equivalently, the difference is exactly $\int_0^1 2x dx = 1$.

Where it goes wrong

- Assuming $I = J$ because the integrands 'look the same'.
- Arithmetic giving $J - I = \frac{4}{3}$ - an offered option.

Takeaway $(x + 1)^2 = x^2 + 2x + 1$: the difference between the two integrals is the integral of the cross term.

Question 2 INT-20 · Integration · spec MM7.2

Evaluate $\int_{-1}^2 |x| dx$.

Answer: D

Two triangles: $\frac{1}{2} \cdot 1 \cdot 1 + \frac{1}{2} \cdot 2 \cdot 2 = \frac{1}{2} + 2 = \frac{5}{2}$.

Full method - how and why

A modulus integrand. Split at the corner, or - faster - read the areas off as triangles.

1 Note $|x|$ changes formula at $x = 0$, which is inside $[-1, 2]$.

why: Split at every point where the modulus switches branch.

2 On $[-1, 0]$: $|x| = -x$, and the graph is a triangle with base 1 and height 1, area $\frac{1}{2}$.

why: Sketching $y = |x|$ makes the integral a pair of triangles - no antiderivative needed.

3 On $[0, 2]$: $|x| = x$, a triangle with base 2 and height 2, area 2.

why: $\frac{1}{2} \times 2 \times 2 = 2$.

4 Add: $\frac{1}{2} + 2 = \frac{5}{2}$.

why: Both pieces are above the axis, so both count positively.

Where it goes wrong

- Integrating x straight through and getting $\frac{4}{2} - \frac{1}{2} = \frac{3}{2}$ - an offered option.
- Treating the left piece as negative.

Takeaway $|x|$ is never negative: split at the corner, and use triangle areas when the graph is piecewise linear.

Question 3 X2-17 · Integration · spec MM7.5

The trapezium rule with n strips is used to estimate $\int_0^1 \sqrt{x} \, dx$. Which of the following is true for every n ?

Answer: B

$y = \sqrt{x}$ is concave (bends downwards), so every chord lies below the curve and the trapezia underestimate the area.

Full method - how and why

Whether the trapezium rule over- or under-estimates depends on the **concavity** of the curve.

- 1 The trapezium rule replaces the curve on each strip by the straight chord joining the endpoints.
why: Its estimate is the area under those chords.
- 2 $y = \sqrt{x}$ is concave: it rises steeply then flattens, bending downwards.
why: Its second derivative $-\frac{1}{4}x^{-3/2}$ is negative for $x > 0$ - or just picture the shape.
- 3 For a concave curve every chord lies **below** the arc, so each trapezium sits below the true region.
why: That is what concavity means geometrically.
- 4 So the estimate is an underestimate, for every n .
why: The number of strips affects accuracy, not direction.

Where it goes wrong

- Assuming the trapezium rule always overestimates - true only for convex (upward-bending) curves such as $y = x^2$.
- Thinking the parity of n matters - that is Simpson's rule, which needs an even number of strips.
- Answering 'cannot be decided' when the shape settles it.

Takeaway Trapezium rule: convex curve $\Rightarrow$ overestimate, concave curve $\Rightarrow$ underestimate. Sketch and look at the bend.

Question 4 INT-01 · Integration · spec MM7.2

Evaluate $\int_1^4 \frac{3x^2 - 5}{\sqrt{x}} \, dx$.

Answer: F

Integrand $= 3x^{3/2} - 5x^{-1/2}$; integral
 $= \left[\frac{6}{5}x^{5/2} - 10x^{1/2} \right]_1^4 = \left(\frac{6 \cdot 32}{5} - 20 \right) - \left(\frac{6}{5} - 10 \right) = \frac{192}{5} - 20 - \frac{6}{5} + 10 = \frac{186}{5} - 10 = \frac{136}{5}$.

Full method - how and why

You can only integrate powers of x , so split the fraction first. Then the limits are straightforward - if the fractions are handled carefully.

- 1 Split the fraction: $\frac{3x^2 - 5}{\sqrt{x}} = 3x^2x^{-1/2} - 5x^{-1/2} = 3x^{3/2} - 5x^{-1/2}$.
why: Dividing by $\sqrt{x}$ subtracts $\frac{1}{2}$ from each index.
- 2 Integrate: $\int 3x^{3/2} \, dx = \frac{3x^{5/2}}{5/2} = \frac{6}{5}x^{5/2}$ and $\int 5x^{-1/2} \, dx = 10x^{1/2}$.
why: Add one to the index and divide by the new index: $\frac{3}{2} + 1 = \frac{5}{2}$, and $-\frac{1}{2} + 1 = \frac{1}{2}$ so dividing by $\frac{1}{2}$ doubles.
- 3 Evaluate at $x = 4$: $4^{5/2} = 32$ and $4^{1/2} = 2$, giving $\frac{192}{5} - 20$.
why: $4^{5/2} = (\sqrt{4})^5 = 2^5$.
- 4 Evaluate at $x = 1$: $\frac{6}{5} - 10$. Subtract: $\left(\frac{192}{5} - 20 \right) - \left(\frac{6}{5} - 10 \right) = \frac{186}{5} - 10 = \frac{136}{5}$.
why: $\frac{192-6}{5} = \frac{186}{5}$, then $-20 + 10 = -10 = -\frac{50}{5}$.

Where it goes wrong

- Forgetting the lower limit entirely, giving $\frac{186}{5}$ - an offered option.
- Dividing by $\frac{5}{2}$ incorrectly: $\frac{3}{5/2} = \frac{6}{5}$, not $\frac{15}{2}$.

Takeaway Split fractions into powers of x before integrating; $4^{5/2} = 32$ type evaluations are worth doing slowly.

Question 5 INT-05 · Integration · spec MM7.1

Find the area of the finite region enclosed by the curve $y = x^2$ and the line $y = 2x + 3$.

Answer: E

Intersections: $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$. Area
 $= \int_{-1}^3 (2x + 3 - x^2) dx = \left[x^2 + 3x - \frac{x^3}{3} \right]_{-1}^3 = (9 + 9 - 9) - \left(1 - 3 + \frac{1}{3} \right) = 9 + \frac{5}{3} = \frac{32}{3}$.

Full method - how and why

Area between a line and a curve: integrate (upper – lower) between the intersection points.

- 1 Find the intersections: $x^2 = 2x + 3 \Rightarrow x^2 - 2x - 3 = (x - 3)(x + 1) = 0$, so $x = -1$ and $x = 3$.
why: These are the limits of the enclosed region.
- 2 Decide which is on top: at $x = 0$ the line gives 3 and the curve gives 0, so the line is above.
why: Testing one interior point settles the order - guessing wrong gives the negative of the answer.
- 3 Integrate the difference: $\int_{-1}^3 (2x + 3 - x^2) dx = \left[x^2 + 3x - \frac{x^3}{3} \right]_{-1}^3$.
why: Subtracting first, then integrating once, is quicker and less error-prone than two separate integrals.
- 4 Evaluate: at 3, $9 + 9 - 9 = 9$; at -1 , $1 - 3 + \frac{1}{3} = -\frac{5}{3}$. Difference: $9 + \frac{5}{3} = \frac{32}{3}$.
why: $-\frac{(-1)^3}{3} = +\frac{1}{3}$ - the double negative is the classic slip.

Where it goes wrong

- Integrating (curve – line) and giving $-\frac{32}{3}$, or taking the magnitude of the wrong piece.
- Sign error at the lower limit, giving $\frac{16}{3}$ or $\frac{28}{3}$.

Takeaway Area between curves = $\int(\text{upper} - \text{lower})$ between intersections; check which is upper with one test value.

Question 6 INT-11 · Integration · spec MM7.1

The region bounded by $y = 4 - x^2$, the x -axis and the lines $x = 0$ and $x = 3$ is shaded. Find the total shaded area.

Answer: E

From 0 to 2 the curve is above the axis: $\int_0^2 (4 - x^2) dx = 8 - \frac{8}{3} = \frac{16}{3}$. From 2 to 3 it is below:
 $\left| \int_2^3 (4 - x^2) dx \right| = \left| (12 - 9) - \frac{16}{3} \right| = \frac{7}{3}$. Total $\frac{23}{3}$.

Full method - how and why

The curve crosses the axis inside the range, so the region must be split - otherwise the pieces cancel.

- 1 Find where the curve meets the axis: $4 - x^2 = 0$ at $x = \pm 2$, so $x = 2$ is inside $[0, 3]$.
why: A crossing inside the limits is the signal to split the integral.
- 2 From 0 to 2 the curve is above the axis: $\int_0^2 (4 - x^2) dx = \left[4x - \frac{x^3}{3} \right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3}$.
why: Positive, as expected.
- 3 From 2 to 3 it is below: $\int_2^3 (4 - x^2) dx = (12 - 9) - \left(8 - \frac{8}{3} \right) = 3 - \frac{16}{3} = -\frac{7}{3}$.
why: Negative, so take its magnitude $\frac{7}{3}$ for the area.
- 4 Add the magnitudes: $\frac{16}{3} + \frac{7}{3} = \frac{23}{3}$.
why: Total shaded area, both pieces counted positively.

Where it goes wrong

- Integrating straight through from 0 to 3 and getting 3 - which is offered.
- Forgetting to take the magnitude of the second piece.

Takeaway Split at every crossing inside the limits and add the magnitudes; a single integral would cancel the pieces.

Question 7 INT-14 · Integration · spec MM7.5

The trapezium rule is used to estimate $\int_a^b f(x) dx$. For which of the following functions is the estimate guaranteed to be an underestimate for every choice of $a < b$?

Answer: A

The trapezium rule underestimates when the curve is concave (bends downwards) so that chords lie below the curve: $y = -x^2$ everywhere. x^3 changes concavity at 0.

Full method - how and why

A conceptual question about the trapezium rule: it is the curve's concavity, not its sign or steepness, that decides.

1 Recall the rule: the trapezium rule replaces the curve by straight chords.

why: Each strip's true area is compared with a trapezium whose top is the chord.

2 A chord lies **below** a concave (downward-bending) curve, so the estimate is too small: an underestimate.

why: Concave means $f'' < 0$ throughout - the curve bulges above its chords.

3 Check the options: $-x^2$ has $f'' = -2 < 0$ everywhere ✓. x^2 and x^4 are convex (overestimate); $2x + 1$ is straight (exact); x^3 changes concavity at 0, so it depends on the interval.

why: 'Guaranteed for every choice of $a < b$ ' rules out x^3 , whose behaviour varies.

4 So $f(x) = -x^2$.

why: The only universally concave option.

Where it goes wrong

- Choosing x^3 without noticing that it is concave for $x < 0$ but convex for $x > 0$.
- Thinking a negative function must give an underestimate - sign is irrelevant; shape is everything.

Takeaway Trapezium rule: concave ($f'' < 0$) underestimates, convex ($f'' > 0$) overestimates, linear is exact.

Question 8 INT-15 · Integration · spec MM7.1

The area between the curve $y = x^2$ and the line $y = k$ (where $k > 0$) is $\frac{32}{3}$. Find k .

Answer: D

$$\text{Area} = \int_{-\sqrt{k}}^{\sqrt{k}} (k - x^2) dx = 2 \left(k^{3/2} - \frac{k^{3/2}}{3} \right) = \frac{4}{3} k^{3/2} = \frac{32}{3} \Rightarrow k^{3/2} = 8 \Rightarrow k = 4.$$

Full method - how and why

Work backwards from a given area to the parameter. Symmetry halves the work.

1 Find the intersections: $x^2 = k$ at $x = \pm\sqrt{k}$.

why: The horizontal line cuts the parabola at two symmetric points.

2 Set up the area: $\int_{-\sqrt{k}}^{\sqrt{k}} (k - x^2) dx$, with the line above the curve between the roots.

why: Upper minus lower.

3 Use symmetry: $= 2 \int_0^{\sqrt{k}} (k - x^2) dx = 2 \left[kx - \frac{x^3}{3} \right]_0^{\sqrt{k}} = 2 \left(k^{3/2} - \frac{k^{3/2}}{3} \right) = \frac{4}{3} k^{3/2}$.

why: The integrand is even, so integrate over half the range and double. Note $k\sqrt{k} = k^{3/2}$.

4 Solve $\frac{4}{3} k^{3/2} = \frac{32}{3}$: $k^{3/2} = 8$, so $k = 8^{2/3} = 4$.

why: Undo the $\frac{3}{2}$ power by raising to the $\frac{2}{3}$: $8^{2/3} = (\sqrt[3]{8})^2 = 4$.

Where it goes wrong

- Solving $k^{3/2} = 8$ as $k = 8^{3/2} = 22.6$ or $k = \sqrt{8}$ - both offered in spirit.
- Forgetting to double after using symmetry.

Takeaway Use even symmetry to halve the integration; undo a $\frac{3}{2}$ power by raising to $\frac{2}{3}$.

Question 9 INT-19 · Integration · spec MM7.1

The curve $y = x(x - 1)(x - 2)$ and the x -axis enclose two regions. What is the ratio of the area of the region for $0 \leq x \leq 1$ to the area for $1 \leq x \leq 2$?

Answer: E

The cubic has rotational symmetry about $(1, 0)$ (it is $u^3 - u$ with $u = x - 1$, an odd function), so the two regions are congruent: ratio 1 : 1.

Full method - how and why

A symmetry question. Shifting the cubic reveals it is odd about $(1, 0)$, so the two regions are congruent - no integration needed.

1 Substitute $u = x - 1$ to centre the cubic: $y = (u + 1)u(u - 1) = u(u^2 - 1) = u^3 - u$.

why: The roots 0, 1, 2 become $-1, 0, 1$ - symmetric about the origin, which is the hint to try this.

2 $u^3 - u$ is an **odd** function: replacing u by $-u$ negates it.

why: Odd functions have rotational symmetry of order 2 about the origin.

3 So the piece on $-1 \leq u \leq 0$ is the 180° rotation of the piece on $0 \leq u \leq 1$: the regions are congruent and have equal areas.

why: Rotation preserves area.

4 The ratio is 1 : 1.

why: You could integrate both (each has area $\frac{1}{4}$), but the symmetry argument is instant.

Where it goes wrong

- Assuming the regions differ because one is above the axis and one below - sign differs, area does not.
- Integrating carelessly and comparing signed values.

Takeaway Shift a cubic so its middle root is at the origin: if it becomes odd, the two enclosed regions are congruent.

Question 10 X2-03 · Integration · spec MM7.1

f is a function defined for all real x and $\int_0^4 f(x) dx = 0$. Which of the following statements must be true?

- I $f(x) = 0$ for some x with $0 \leq x \leq 4$.
- II The area between the curve and the x -axis from 0 to 4 is 0.
- III $\int_0^2 f(x) dx = -\int_2^4 f(x) dx$

Answer: D

III is just the rule for combining integrals over adjacent ranges. I fails for a function that jumps from -1 to 1 without passing through 0 (nothing says f is continuous). II fails: $f(x) = x - 2$ has integral 0 but area 4.

Full method - how and why

A hard 'must be true' about integrals. Two claims import assumptions the question never makes.

- 1 III: $\int_0^4 = \int_0^2 + \int_2^4$, and the left side is 0, so $\int_0^2 = -\int_2^4$. **True** - pure algebra of integrals.

why: Splitting an integral at an interior point is always valid where the integral exists.

- 2 I: 'zero integral $\Rightarrow f$ hits zero' is the intermediate value theorem in disguise - but that needs **continuity**, which is not given.

why: Read the hypotheses: it only says f is defined for all real x .

- 3 Counterexample for I: a step function equal to -1 on $[0, 2)$ and 1 on $[2, 4]$ has integral 0 but never equals 0.

False.

why: It jumps straight past zero.

- 4 II: 'integral 0' means signed areas cancel, not that the area is 0. Take $f(x) = x - 2$: $\int_0^4 (x - 2) dx = 0$, but the area is $2 + 2 = 4$. **False.**

why: Area counts the region below the axis as positive; the integral counts it as negative.

- 5 So III only.

why: Match to the list.

Where it goes wrong

- Accepting I by silently assuming f is continuous - an assumption the question deliberately withholds.
- Treating 'integral' and 'area' as the same thing; they agree only when $f \geq 0$ throughout.

Takeaway Integral $\neq$ area when the function changes sign; and the intermediate value theorem requires continuity, which must be **given**.